Reg. No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION ….

Third Semester

MODEL QUESTION PAPER - THEORY

Minor Course - MG3DSCSTA202–

DATA ANALYSIS IN INFERENTIAL STATISTICS USING R/PYTHON

(2024 ADMISSION)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Short Answer Questions

*Answer any **seven** questions. Each question carries **2 marks**.*

1. Given the joint probability mass function, $f(x, y) = c(x+y)$, defined over the points (1,1), (2,1), (2,2), (3,1), determine the value of the constant c. [U][1]
2. What is the definition of a conditional probability density function (pdf)? [U][1]
3. State the Central Limit Theorem. [U][2]
4. If X follows a normal distribution with mean 2 and standard deviation 3, find the probability that the sample mean $\bar{X}$ is greater than 2.5, given a sample size of 16. [U][2]
5. State the sufficient conditions for consistency of a parameter. [K][3]
6. Obtain the moment estimate for μ in N (μ, σ), σ known. [A][3]
7. Briefly explain the two types of errors that can occur in statistical hypothesis testing. [U][4]
8. A sample of 100 students is taken from a population with heights having standard deviation 10 cm. The mean height of the sample of students is found to be 168.8cm. Can we accept the assumption that the mean height of the students is 170 cm? [A][4]
9. State any two conditions required for the valid application of the non-parametric Chi-square tests. [K][4]
10. A sample of 10 individuals is chosen at random from a normal population and their mean weight is found to be 67.8 Kg with standard deviation 2.826 Kg. Check whether the mean of the population is 66. [A][4]

(7x2=14 marks)

Part B

Short Essay Questions

*Answer any **four** questions. Each question carries **6 marks**.*

11. For (X, Y), the joint *pdf* is given by $f(x, y) = 30xy^2$; $x = 1, 2, 3$; $y = 1, 2$. Check independence of X and Y. [U][1]
12. Define Student's *t* distribution. State any two of its properties. Mention any one Student's *t* statistic. [U][2]
13. Explain the concepts of efficiency and sufficiency as desirable properties of point estimators, and support your explanation with one example each. [U][3]
14. Obtain maximum likelihood estimators of the parameters of uniform distribution over the interval (a, b). [A][3]
15. The average wage of a sample of 150 workers in factory A was Rs. 2.56 with standard deviation Rs. 1.08. A sample of 200 workers in factory B has average hourly wage of Rs. 2.87 with standard deviation Rs. 1.28. Can the applicant safely assume that the hourly wage paid by factory B is higher than that of factory A? [An][4]
16. Briefly discuss ANOVA. If the Between Sum of Squares (SSC) is 123.33 and the Within Sum of Squares (SSE) is 94 for data collected on the test scores of students taught using three different teaching methods, conduct a one-way ANOVA to test whether there is a significant difference in the average scores among the teaching methods. Assume that the data was collected from 5 students per group (total of 15 students), and use a 5% significance level for the test. [An][4]

(4x6=24 marks)

Part C

Essay Type Questions

*Answer any **one** question. Each question carries **12 marks**.*

17. Define chi-square distribution and state any four of its properties. For a sample of size 10 from a normal population with standard deviation 5, find $P(8.4 < S^2 < 42.3)$. [U][2]
18. Fit a Poisson distribution to the following frequency data and perform a goodness-of-fit test. [An][4]

x	0	1	2	3	4
f	17167	1861	124	2	1

(1x12=12 marks)